Supplementary Material

S2 — COREQ (Consolidated Criteria for Reporting Qualitative Research) Checklist

Paper Title: *Explainable Machine Learning for Student Burnout Classification and Risk Stratification: A Mixed-Methods Study with Qualitative Triangulation*

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32-item checklist — Tong, Sainsbury & Craig (2007)

Domain 1: Research Team and Reflexivity

Personal Characteristics

No.	Item	Guide Questions	Response
1	Interviewer / Facilitator	Which author(s) conducted the interview?	The primary author (Rifat Miah) conducted all 20 semi-structured interviews.
2	Credentials	What were the researcher's credentials?	BSc in Computer Science and Engineering (enrolled at Presidency University at time of data collection and manuscript submission).
3	Occupation	What was their occupation at the time of the study?	Undergraduate researcher and ML practitioner.
4	Gender	Was the researcher male or female?	Male
5	Experience and training	What experience or training did the researcher have?	Completed coursework in research methodology; reviewed Braun & Clarke (2006, 2021) thematic analysis protocols prior to fieldwork. Interview guide reviewed and piloted before use.

Relationship with Participants

No.	Item	Guide Questions	Response
6	Relationship established	Was a relationship established prior to study commencement?	Yes. The primary researcher was a peer student at participating institutions; some participants were known via institutional networks. Acknowledged as a reflexivity consideration in Section 8.2.
7	Participant knowledge of the interviewer	What did participants know about the researcher?	All participants were informed via the written informed consent form of: (i) the academic purpose, (ii) the researcher's identity and affiliation, (iii) the voluntary nature of participation, and (iv) intended use for academic publication.

No.	Item	Guide Questions	Response
8	Interviewer characteristics	What characteristics were reported about the interviewer?	Bengali-speaking male student from a Bangladeshi private university. Reflexivity mitigation: qualitative open-coding completed before reviewing SHAP feature rankings; independent second coder employed (κ = 0.82, Section 3.8).

Domain 2: Study Design

Theoretical Framework

No.	Item	Guide Questions	Response
9	Methodological orientation and theory	What methodological orientation was stated to underpin the study?	Reflexive Thematic Analysis (Braun & Clarke, 2006, 2021) within an Explanatory Sequential Mixed-Methods Design (QUAN → QUAL; Creswell & Plano Clark, 2018). Theoretical frameworks: COR Theory, JD-R Model, Self-Determination Theory.

Participant Selection

No.	Item	Guide Questions	Response
10	Sampling	How were participants selected?	Nested purposive maximum variation sampling. N = 20 selected from N = 601 cohort across three burnout severity strata (Low: n = 2, Medium: n = 8, High: n = 10), institutional type, academic year, and degree level.
11	Method of approach	How were participants approached?	Via direct message, WhatsApp, or email. Approached participants who had previously indicated willingness for follow-up contact in the quantitative survey. All 20 consented participants were successfully interviewed.
12	Sample size	How many participants were in the study?	N = 20 qualitative interview participants.
13	Non-participation	How many people refused to participate or dropped out?	Three initially willing participants declined to schedule (citing time / academic pressure); replaced by next eligible stratum members. No mid-interview withdrawals.

Setting

No.	Item	Guide Questions	Response
14	Setting of data collection	Where was the data collected?	In-person: quiet campus spaces (library study rooms, cafeteria areas). Remote: WhatsApp audio calls or Google Meet, as preferred by participant.

No.	Item	Guide Questions	Response
15	Presence of non-participants	Was anyone else present besides the participants and researchers?	No. All interviews were one-on-one and confidential.
16	Description of sample	What are the important characteristics of the sample?	See Table 8 (Section 8) for full demographic characteristics. Participants spanned all academic years (1st–Final), multiple degree programs (Bachelor's, Master's), and three burnout tiers (Low n=2, Medium n=8, High n=10). Semi-structured qualitative interviews were conducted during June–July 2026, following completion of the quantitative survey data collection phase (completed May 2026; survey conducted March–May 2026).

Data Collection

No.	Item	Guide Questions	Response
17	Interview guide	Were questions, prompts, guides provided by the authors? Was it pilot tested?	Yes. Structured 10-item interview guide used covering: daily study routine, sleep duration, social media, part-time employment, academic stress, mental health, physical activity, subjective burnout, attendance, sleep quality. Reviewed by co-author; pilot tested with 2 volunteer students not in main sample.
18	Repeat interviews	Were repeat interviews carried out?	No. Each participant was interviewed once.
19	Audio/visual recording	Did the research use audio or visual recording?	Yes. All interviews were audio-recorded with participant consent, transcribed verbatim in Bangla, and translated into English with back-translation verification.
20	Field notes	Were field notes made during and/or after the interview?	Yes. The primary interviewer maintained reflective notes after each session documenting contextual observations, emerging patterns, and reflexive considerations.
21	Duration	What was the duration of the interviews?	Approximately 20–30 minutes per interview (range: 18–35 minutes).
22	Data saturation	Was data saturation discussed?	Yes. Thematic saturation assessed iteratively; no new major themes emerged after the 15th interview. Final 5 interviews confirmed redundancy across all four themes. Consistent with Guest et al. (2006) benchmarks for homogenous populations.
23	Transcripts returned	Were transcripts returned to participants for comment and/or correction?	Participants were offered transcript review; none requested corrections or changes.

Domain 3: Analysis and Findings

Data Analysis

No.	Item	Guide Questions	Response
24	Number of data coders	How many data coders coded the data?	Two coders: (1) primary author coded all 20 transcripts; (2) independent bilingual second coder coded 25% random sub-sample (n = 5 transcripts).
25	Description of the coding tree	Did authors provide a description of the coding tree?	Yes. Four overarching themes with sub-themes identified and reported in Sections 8.3–8.6: (1) Academic Dread and Career Despair; (2) Institutional Identity and Low-Pressure Burnout; (3) Digital Fatigue as Maladaptive Coping; (4) Circadian Collapse and Sleep Deprivation.
26	Derivation of themes	Were themes identified in advance or derived from the data?	Themes derived inductively from the data (bottom-up open coding), completed before reviewing quantitative SHAP feature rankings. Subsequently mapped onto JD-R and COR frameworks in the triangulation matrix (Table 9, Section 9).
27	Software	What software was used to manage the data?	Manual coding using structured codebook in Microsoft Word/Excel. No qualitative data management software used; corpus size (N = 20 transcripts × 10 questions) was manageable without software.
28	Participant checking	Did participants provide feedback on the findings?	Not formally completed. The independent inter-rater reliability verification (κ = 0.82, Section 3.8) serves as a methodological proxy for analytical trustworthiness.

Reporting

No.	Item	Guide Questions	Response
29	Quotations presented	Were participant quotations presented to illustrate the themes/findings?	Yes. Representative verbatim quotations provided in Sections 8.3–8.6 for all four themes. Each quotation accompanied by participant identifier (P1–P20), degree program, academic year, institutional type, and burnout tier.
30	Data and findings consistent	Was there consistency between the data presented and the findings?	Yes. Themes are consistently grounded in verbatim quotations. Triangulation matrix (Table 9, Section 9) explicitly maps SHAP signals to qualitative themes, confirming cross-strand consistency.
31	Clarity of major themes	Were major themes clearly presented in the findings?	Yes. Four major themes presented with dedicated subsections (Sections 8.3–8.6), each with thematic title, descriptive narrative, representative quotations with participant identifiers, and methodological caveats where applicable.
32	Clarity of minor themes	Is there a description of diverse cases or discussion of minor themes?	Yes. Theme 2 (Institutional Identity Burnout) noted as emerging from n = 2 National University participants and framed as exploratory/hypothesis-generating (Section 8.4). Protective patterns (e.g., P5 as resilient

No.	Item	Guide Questions	Response
			low-burnout case) contrasted with high-burnout cases in Section 8.6.

COREQ Compliance Summary

Domain	Items Reported	Compliance
Domain 1: Research Team and Reflexivity	8/8	**Full**
Domain 2: Study Design	14/14	**Full**
Domain 3: Analysis and Findings	9/10	Full (Item 28 partially met via IRR as proxy)
Total	**31/32**	**97% Compliance**

Adapted from: Tong, A., Sainsbury, P., & Craig, J. (2007). Consolidated criteria for reporting qualitative research (COREQ): A 32-item checklist for interviews and focus groups. International Journal for Quality in Health Care, 19(6), 349–357. https://doi.org/10.1093/intqhc/mzm042